



C.3 Wave phenomena

Wave behaviour · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Reflection and refraction occur at boundaries between media; refraction follows Snell's law, $n_1 \sin \theta_1 = n_2 \sin \theta_2$, where n is refractive index, and can cause total internal reflection when travelling from higher to lower refractive index beyond the critical angle, $\sin \theta_c = n_2/n_1$. Diffraction is the spreading of waves around obstacles or through apertures, most significant when aperture size is comparable to wavelength; single-slit diffraction produces a central maximum with minima at $\sin \theta = m\lambda/b$ ($m = 1, 2, 3, \dots$). Interference occurs when coherent waves superpose: double-slit interference gives bright fringes at path difference $d \sin \theta = m\lambda$, with fringe spacing $s = \lambda D/d$ for slit separation d and screen distance D ; diffraction gratings produce sharp maxima at $d \sin \theta = m\lambda$ with much narrower, brighter fringes than a double slit.

Worked example: follow the reasoning

Light of wavelength 600 nm passes through a double slit with slit separation 0.20 mm, forming fringes on a screen 2.0 m away. Find the fringe spacing.

$$s = \lambda D/d = (600 \times 10^{-9} \times 2.0)/(0.20 \times 10^{-3}) = (1.2 \times 10^{-6})/(2.0 \times 10^{-4}) = 6.0 \times 10^{-3} \text{ m.}$$

Answer: $s = 6.0 \text{ mm}$.

Quick check · multiple choice

Light travels from glass ($n = 1.5$) into air ($n = 1.0$) at an angle greater than the critical angle. What happens to the light?

- A It refracts into the air, bending away from the normal
- B It refracts into the air, bending towards the normal
- C It undergoes total internal reflection
- D It is completely absorbed

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Monochromatic light of wavelength 550 nm is incident normally on a diffraction grating with 500 lines per mm. [7]
- (a) Calculate the grating spacing d .
 - (b) Calculate the angle of the first-order ($m=1$) maximum.
 - (c) Determine the highest order of maximum that can be observed for this grating and wavelength.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $d = 1/(500 \times 10^3 \text{ lines/m}) = 1/(5.0 \times 10^5) = 2.0 \times 10^{-6} \text{ m}$.

(b) $d \sin \theta = m \lambda \rightarrow \sin \theta = (1)(550 \times 10^{-9})/(2.0 \times 10^{-6}) = 0.275 \rightarrow \theta = 16.0^\circ$.

(c) Maximum order when $\sin \theta \leq 1$: $m \leq d/\lambda = (2.0 \times 10^{-6})/(550 \times 10^{-9}) = 3.64$, so highest integer order is $m = 3$.

Quick check answer

C. Beyond the critical angle when going from a higher to a lower refractive index medium, all the light is totally internally reflected back into the denser medium.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.